Math 300: Introduction to Mathematical Reasoning

Spring 2025

Course information

Times:	MWF 10:30–11:20 a.m.
Location:	205 Loew Hall.
Web page:	**https://canvas.uw.edu/courses/1801533**

Instructors information

Instructor:	Guillermo Sanmarco (he/him).
Email:	**sanmarco@uw.edu**
Office:	C-437 Padelford Hall.
Office hours:	Wednesday 12:00–2:00 p.m.

TA:	Natasha Crepeau.
Email:	**ncrepeau@uw.edu**
Office:	C-446 Padelford Hall.
Office hours:	Tuesday 12:00–1:30 p.m.

Course format

The course will consist of blackboard lectures. There will be 8 homework assignments, a midterm and a final exam (details below). Assignments will be submitted and graded with Gradescope. You can post questions to our class Piazza.

Students are highly encouraged to take notes and participate during class. The topics to be discussed in each lecture, and the corresponding section in the main textbook, will be published in advance, see the "Schedule and lecture notes" Canvas module. Lecture notes will be posted after each session.

Textbooks

- **Main reference:** Mathematical Reasoning: Writing and Proof, Version 3, by Ted Sundstrom. A free pdf has been made available by the author **here.**
- **Secondary reference:** An Introduction to Mathematical Reasoning, by Matthew Conroy and Jennifer Taggart. Available for free **here.**

We will cover most of the main textbook. Reading the sections of the textbook corresponding to the assigned homework exercises is considered part of the assignment.

Course topics

In this course we will learn some of the most fundamental tools of mathematics: logic, proof techniques, set theory, cardinality, functions, etc. You will use these tools and concepts in every abstract mathematics course you take.

Learning how to elaborate and communicate mathematical arguments is the most important skill to develop in this course. This entails getting comfortable with abstract reasoning, which is not

necessarily how we operate in a day-to-day basis. The good news: if you work hard in this course, your reasoning capacities will change for the better! Another fundamental skill for abstract math that we will develop in this class is how to problem solve and think creatively.

Homework

You will have 8 weekly homework assignments that will be submitted and graded on Gradescope. Assignments will be due on **Friday by 10:00 am**. Late assignments will not be accepted. Your scores will be averaged to compute your final homework score.

Inevitably, you will mistakes as you learn how to construct proofs. At the end of the term, I will add an extra 5% to your total homework score, unless you are already in the 95%–100% range.

Getting help: I expect you to attend office hours or contact me by e-mail if you need help with homework assignments or anything else in this course.
Collaborative work and study groups are strongly encouraged. These are great tools to help you polish your own ideas and also to take a glimpse at lines of thought that you might not have considered. You are allowed to discuss graded assignments outside of class. However, the grade that you receive on an assignment should represent your understanding, not someone else's. You are not allowed to take written or electronic records away from group sessions. You should wait until you are on your own to write down your solutions.

Exams

This course will have two exams: a midterm and a final. Topics covered in the midterms may be also tested on the final. To pass the final, you will need to understand all the material covered in the course.

The **midterm** will take place on **Monday, April 28** during the regular class location and time. The final exam is scheduled for **Monday, June 9 from 8:30 to 10:20 am.**

Grading policy

Homework 25%
Midterm 35%
Final 40%

A word of advice

A first proof-oriented math course can be quite a challenge. The process of learning how to come up with ideas and communicate them effectively is not always a smooth one. You must have the courage and self-discipline to keep trying even when you feel frustrated. Keep in mind that feeling frustrated is completely normal, and talking about it usually helps. My advice for you is to stay on top of assignments, assist office hours, and try to socialize with your peers. I feel that following a textbook can be very anchoring for students. I highly recommend reading the material before class. I don't pretend you to understand even 10% of what you read on your own, but you will notice that doing so will help you ask questions during class, which I really appreciate.

Your instructor cares about your well-being

You may experience stressors that can impact both your academic experience and your personal well-being. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.

If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. Asking for support sooner rather than later is often helpful. Many helpful resources are available on campus, and an important part of the college experience is learning how to ask for help. If you encounter stressors of any kind, please contact me so that we can find solutions together.

Excused absence from class

Students are expected to attend class and to participate in all graded activities, including midterms and final examinations. A student who is anticipating being absent from class due to a Religious Accommodation activity needs to complete the Religious Accommodations request process by the second Friday of the quarter. Students who anticipate missing class due to attendance at academic conferences or field trips, or participation in university-sponsored activities should provide a written notice to the instructor ahead of the absence. The instructor will determine if the graded activity or exam can be rescheduled or if there is equivalent work that can be done as an equivalent, as determined by the instructor.

Medical excuse notes

Students are expected to attend class and to participate in all graded activities, including midterms and final examinations. To protect student privacy and the integrity of the academic experience, students will not be required to provide a medical excuse note to justify an absence from class due to illness. A student absent from any graded class activity or examination due to illness must request, in writing, to take a rescheduled examination or perform work judged by the instructor to be the equivalent. Students are responsible for taking any number of examinations for which they are scheduled on a given day and may not request an adjustment for this reason alone.

Religious accommodation

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Religious Accomodations Policy. Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form.